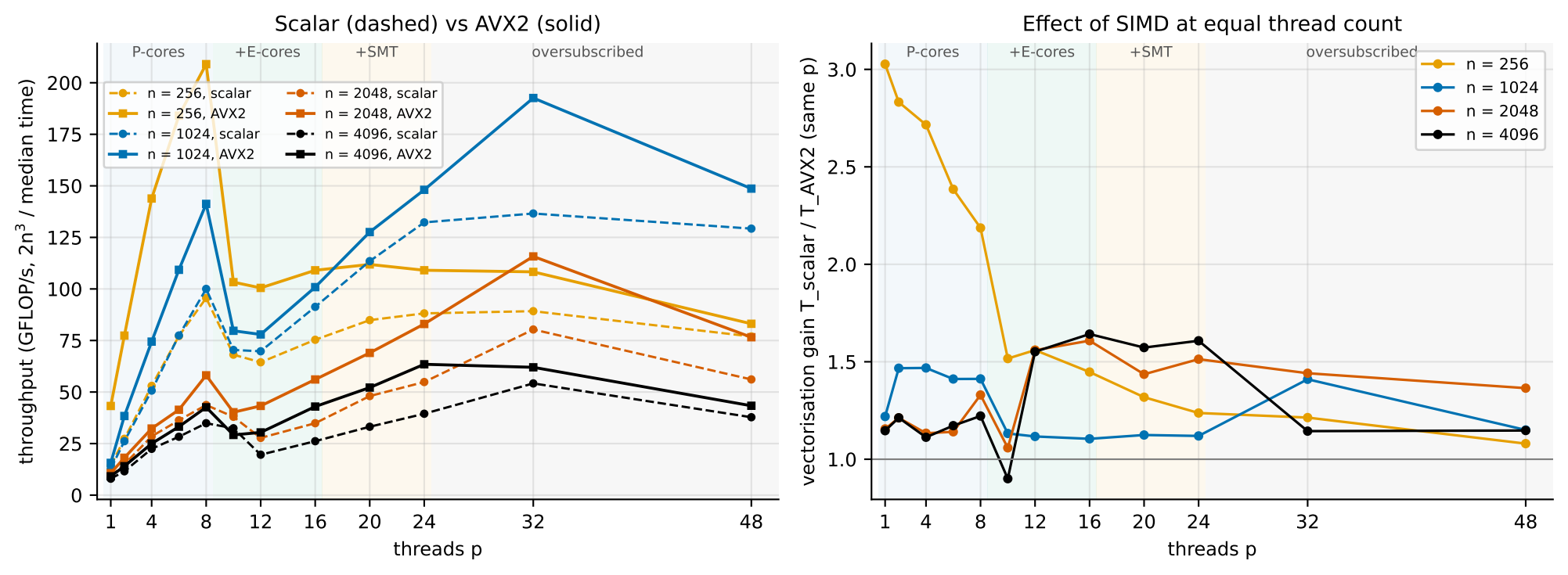
Scalar (dashed) vs AVX2 (solid)
P-cores
+E-cores
+SMT
oversubscribed
n = 256, scalar
n = 256, AVX2
n = 1024, scalar
n = 1024, AVX2
n = 2048, scalar
n = 2048, AVX2
n = 4096, scalar
n = 4096, AVX2
throughput (GFLOP/s, 2n³ / median time)
threads p
Effect of SIMD at equal thread count
vectorisation gain T_scalar / T_AVX2 (same p)
n = 256
n = 1024
n = 2048
n = 4096